Antier Healthcare Suite – Product Build Spec (Sandbox v1)

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Purpose and positioning

Build a working sandbox environment of **Antier Healthcare Suite**: a product sandbox with sample data, not a prototype made for one client. It will be shown live to hospital prospects, first to a Quality Manager at a private hospital in Oman, in a 15-minute walkthrough.

The buyer cares most about two things: patient feedback after discharge, and quality management (incidents, infection-control audits, accreditation tracking, KPIs). Those two modules get the depth. Patient enquiry on WhatsApp is a light front-end flow.

Rules that keep it a product:

- Branding is **Antier Healthcare Suite** everywhere: login, header, emails, PDF exports. No client name or logo anywhere in the build, code, seed data or URLs.
- The tenant in the data is **Demo Hospital**. Every patient, doctor and staff name is fictional. The header carries a small **Sandbox** badge, the way SaaS products mark a test environment; the words "demo", "POC" and "prototype" never appear.
- Every screen a real customer would expect exists and works: admin, users and roles, notifications, audit trail, integrations, reports, help. The next section lists them. No menu item may lead to a dead end.
- It must feel configurable: departments, KPIs, accreditation standards, feedback questions, SLAs, branding and languages are edited in Settings, not hard-coded.
- Real AI where it matters (sentiment, classification, summaries, Arabic). External systems (HIS, WhatsApp, Google) run through sandbox connectors on the Integrations page.
- Nothing on screen claims a live customer, a client logo, or real outcome figures.

Product shell: what makes it real software

A prospect judges maturity by the plumbing around the features: admin, permissions, notifications, audit trail, reports. Their absence is what gives a demo away, so they are in scope.

Area	What to build	Priority
Navigation	Left sidebar: Home, Patient Experience (Feedback, Cases, Enquiries), Quality (Incidents, Audits, Accreditation, Documents, Risks), Reports, Assistant, Admin. Breadcrumbs; global search across patients, cases, incidents and documents	Must

Area	What to build	Priority
Organisation setup	Hospital profile, sites (site switcher, one site populated), departments, wards	Must
Users and roles	Invite, assign role and departments, deactivate; permission matrix per role (view, create, approve, admin); users see only their departments' records unless granted more	Must
Notifications	In-app notification centre plus email templates: assignment, SLA breach, overdue corrective action, audit due, document review due; escalation rules (for example SLA breach → Quality Manager)	Must
Audit trail	Searchable log screen: who, what, when, before and after; filter by user, module and record; export	Must
Integrations	Connector cards: HIS (HL7 v2 / FHIR), WhatsApp Business, Google Business Profile, SMTP email, SMS gateway, SSO (SAML / Azure AD). HIS and WhatsApp show "Sandbox", the rest "Available". The HIS card opens an event console that sends a sample discharge	Must
Reports	Saved reports (monthly quality pack, complaints summary, incident summary, audit compliance); date-range and department filters; PDF and Excel export; scheduled email delivery	Must

Area	What to build	Priority
Patient-facing branding	Feedback form and messages use the hospital's logo, colours and sender name from Settings	Must
UX quality	Pagination, sortable tables, saved filters, bulk assign and close, guided empty states, loading skeletons, toasts, confirmation dialogs, form validation messages	Must
Data import	CSV import for staff, departments and historic complaints and incidents, with column mapping and an error report	Should
Privacy and data	Data-retention settings, patient-name masking toggle, consent line on the feedback form, deployment note (hospital server or approved cloud region)	Should
Account and help	Profile, password change, language switch (EN / AR), timezone; Help centre with short articles; Contact support form	Should

Walkthrough storyline

The build is judged by whether this click path runs end to end without a single dead click, in about 15 minutes. Every screen named here must exist; anything not on this path is optional.

1. **Log in** as Quality Manager (Demo Hospital). Land on the Quality Overview dashboard. *(1 min)*
2. **A patient is discharged.** From Admin → Integrations → HIS (Sandbox), send a discharge event for a patient. A WhatsApp-style feedback message appears in the phone preview panel, in English or Arabic. *(2 min)*
3. **Patient answers.** In the phone preview, submit a 5-star happy response. The app shows the "Would you share this on Google?" step with a review link (simulated). *(1 min)*
4. **A second patient complains.** Submit a 2-star Arabic free-text complaint about billing and waiting time. *(1 min)*

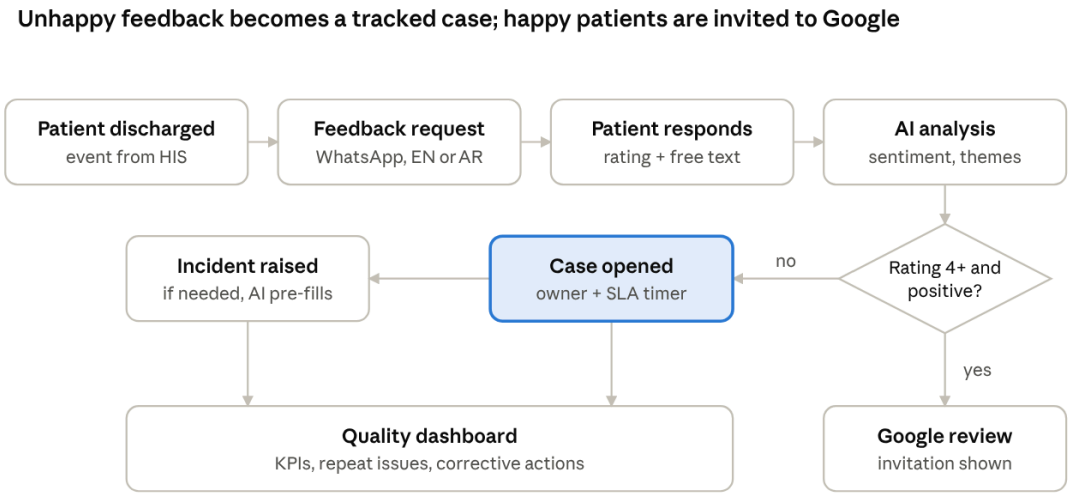
5. **AI reads it.** Open the Feedback inbox: the complaint shows sentiment, themes (Billing, Waiting time), urgency, an English translation, a one-line summary and a suggested owner. A case is opened automatically. *(3 min)*
6. **Complaint becomes an incident.** From the case, press "Raise incident". The incident form is pre-filled by AI; the user edits and submits. Show the incident register and the investigation steps. *(2 min)*
7. **Quality view.** Open the Infection-control audit checklist (one completed, one overdue), then the Accreditation tracker with standards, evidence attached and % readiness. *(3 min)*
8. **KPIs.** Back to the dashboard: complaint closure time, satisfaction trend, incidents by category, audit compliance. Ask the AI assistant "What are the top 3 repeat issues this month?" and get an answer with links to cases. *(2 min)*
9. **Admin (1 min): show Integrations, Users and roles, the Audit trail, and that departments, KPI definitions, accreditation standard sets and feedback questions are editable in Settings**. This is the "we configure it to your hospital" moment.

Module 1: Post-discharge feedback

Every discharged patient gets a short feedback request; happy patients are invited to post a Google review, unhappy ones become an internal case within minutes.

Flow

1. Discharge event arrives from the HIS connector (in the sandbox, sent from the HIS event console) → feedback request queued with a configurable delay (default 24 h; a "Send now" option in the sandbox).
2. Message sent on a simulated WhatsApp channel, in the patient's preferred language (English or Arabic, right-to-left rendering for Arabic).
3. Patient opens a mobile web form: star rating (1–5), 3–5 configurable questions (doctor, nursing, waiting time, billing, cleanliness), optional free text, optional voice note placeholder (UI only).
4. Routing rule (configurable in Settings): rating ≥ 4 **and** AI sentiment positive → show "Share on Google" with a review link; anything else → thank-you screen and an internal case is opened.
5. Never block or filter a patient from leaving a public review. The Google step is an invitation shown to satisfied patients only; unhappy patients still see a neutral thank-you, not a dead end.



feedback routing · 7 steps, 1 decision

Every path ends on the Quality dashboard; only satisfied patients see the Google invitation, and nothing is blocked from posting publicly.

Screens

Screen	What it shows	Key actions
Phone preview panel	WhatsApp-style chat and the mobile feedback form	Fill and submit as the patient
Feedback inbox	All responses, newest first; filters by department, rating, sentiment, theme, language	Open, assign, mark reviewed
Feedback detail	Original text, English translation, sentiment, themes, urgency, AI summary, suggested owner	Accept or edit AI tags, open case, raise incident
Case view	Owner, due date (SLA), status, notes, timeline of every action	Reassign, add note, close with resolution
Review requests	Patients invited to Google, clicked or not (simulated)	None (read-only)

Case statuses: New → Assigned → In investigation → Resolved → Closed. SLA defaults: high urgency 24 h, medium 72 h, low 7 days, all editable in Settings. Overdue cases turn red on the dashboard.

Module 2: Quality management

This is the Quality Manager's home: one place for incidents, infection-control audits, accreditation readiness and KPIs, replacing paper and Google Forms.

2a. Incident management

- Report form usable by any staff role: date and time, department, location, category (patient fall, medication error, infection, equipment, complaint-related, other), severity (near miss, no harm, minor, moderate, severe), description, people involved (staff roles only), immediate action taken.
- AI assist: from free text, suggest category, severity and a one-line summary; the reporter confirms. When raised from a complaint, pre-fill from that complaint.
- Register: table with filters, colour by severity, overdue flag.
- Investigation: assign investigator, root-cause section (5-Whys template), corrective and preventive actions (CAPA) with owner and due date, closure sign-off.
- Link incidents to complaints and to audit findings; show the links both ways.

2b. Infection-control audits

- Audit templates (seed 3): Hand hygiene, Environmental cleaning, Sharps and waste. Each is a checklist of 10–15 items: Yes / No / Not applicable + comment + photo placeholder.
- Schedule: audits by department and due date; statuses Scheduled, Completed, Overdue.
- On submit: compliance % calculated; any "No" can create a corrective action in one click.
- Templates editable in Settings (add item, reorder, change scoring).

2c. Accreditation tracker

- Standard sets loaded as data, not code. Seed one generic set, "Hospital Accreditation Standards (sample)", with 6 chapters and 3–5 standards each (patient rights, infection control, medication management, quality and patient safety, facility safety, governance).
- Do not copy JCI or Oman Ministry standard text. Use our own sample wording; label the set "sample".
- Per standard: owner, status (Not started, In progress, Compliant, Gap), evidence files attached (upload real files in the demo), last reviewed date, notes.
- Readiness view: % compliant per chapter and overall, gaps list, upcoming review dates.

2d. KPI dashboard

KPI	Source in the demo
Patient satisfaction (average rating, trend by month)	Feedback
Complaints opened vs closed, average closure time	Cases
Cases past SLA	Cases
Incidents by category and severity, trend	Incidents
Open corrective actions, overdue CAPA	Incidents, audits
Infection-control audit compliance % by department	Audits

KPI	Source in the demo
Accreditation readiness %	Accreditation tracker
Top repeat themes (AI)	Feedback + incidents

KPI definitions (name, formula, target, red/amber/green thresholds) are records in Settings. Export dashboard and registers to PDF and Excel, branded Antier Healthcare Suite.

2e. Document control (should-have)

- SOP and policy library by department: title, document number, version, owner, effective date, next review date, status (Draft, In review, Approved, Retired).
- Upload a new version; earlier versions kept; approval by the Quality Manager; staff "read and understood" acknowledgement with completion %.
- Overdue reviews appear on the dashboard; documents can be attached as accreditation evidence.

2f. Risk register (should-have)

- Risks by department: description, likelihood × impact score (5 × 5), owner, existing controls, actions, review date, status.
- Heat-map view; links from risks to incidents and to accreditation standards.

Module 3: Patient enquiry on WhatsApp (light)

Keep this to one flow and two screens; it is supporting context, not a lead feature. The prospect already runs a booking chatbot, so this must not look like a replacement for it.

- Phone preview: a patient writes in English or Arabic ("I need an appointment with a cardiologist next week, I have Dhamani insurance"). The AI replies, identifies department and intent (appointment, price question, report request, complaint), and collects name, phone and preferred time.
- If the message is a complaint, it goes straight into the Module 1 feedback inbox.
- Enquiries list: intent, department, language, status (New, Handed to front desk, Closed). A "Hand to front desk" button marks it done; no real booking is made.
- Show a line in the UI: "Booking happens in your HIS". No calendar, no slots, no HIS write.

AI components

Five AI calls are real and run live in the demo; everything else is ordinary application logic.

AI function	Input	Output (JSON)	Where used
Feedback analysis	Rating, answers, free text (EN or AR)	sentiment (positive / neutral / negative + score), themes (from the configured list), urgency (high /	Module 1

AI function	Input	Output (JSON)	Where used
		medium / low), English translation, one-line summary, suggested owner department	
Incident assist	Free-text incident description, or a complaint	category, severity, summary, suggested immediate action	Module 2a
Enquiry triage	WhatsApp message	intent, department, language, extracted fields, reply text in the patient's language	Module 3
Repeat-issue finder	Last 90 days of feedback and incidents	top themes with counts and linked record ids	Dashboard
Quality assistant (Q&A)	A question + the hospital's records and SOP documents	Answer with links to the records or document sections it used; says "not found" when it does not know	Dashboard

Rules:

- Use one LLM provider behind a single adapter so the model can be swapped (we will be asked about on-premise later). Default: a hosted model with strong Arabic.
- Every AI output is a **suggestion** shown with an "AI suggested" label; the user can accept or edit, and both values are stored for the audit trail.
- Structured outputs only (JSON schema), validated; on failure fall back to "Needs review" rather than crashing the demo.
- Themes come from the configurable list in Settings; the model may propose a new theme, flagged as "new".
- No clinical advice of any kind, and the assistant refuses clinical questions politely.
- Pre-compute AI fields for all seed records so the inbox is full on load; only new demo submissions call the model live. Target response under 5 seconds, with a visible "Analysing…" state.

Sample data

The sandbox opens with six months of believable history so every chart and list looks lived-in. All of it is generated and fictional; a seed script must rebuild it from scratch in one command.

Entity	Volume	Notes
Hospital	1: Demo Hospital	~120 beds, private, Gulf region; no city named on screen
Departments	10	OPD, Emergency, Inpatient wards, ICU, Maternity, Paediatrics, Radiology, Laboratory, Billing and insurance, Front office
Users	8	One per role (below); password-protected demo logins
Patients	~600	Gulf, South Asian and Western names; 40% prefer Arabic
Feedback responses	~450 over 6 months	~70% positive, 20% neutral, 10% negative; ~35% with free text, a quarter of those in Arabic
Complaint cases	~60	Mixed statuses; 5–6 overdue at any time
Incidents	~45	All categories and severities; 8 still open, 3 linked to complaints
Infection-control audits	~30	3 templates across departments; compliance 72–96%; 2 overdue
Accreditation standards	~25 in 6 chapters	Readiness about 68% overall, 4 open gaps
WhatsApp enquiries	~80	Mostly appointments and price questions; Dhamani and other insurer names allowed as generic text
SOP documents (for the assistant)	4–5 short PDFs we write	Complaint handling, incident reporting, hand hygiene, feedback policy

Roles: Quality Manager (sees everything), Patient Relations Officer, Department Head, Infection Control Nurse, Staff reporter, Front desk, Management viewer (read-only), Admin (Settings).

Keep satisfaction scores and trends plausible, not perfect: some dips, one bad month for waiting-time complaints, an improving trend after a corrective action. That gives the presenter something to point at.

Tech stack and hosting

A single web app on a stable public URL, built so it can later run on a hospital's own server. The stack below is a recommendation; the developer may swap like for like.

Layer	Recommendation
Front end	Next.js + React, Tailwind, a clean component library (shadcn/ui); responsive; RTL support for Arabic screens
Back end	Node (NestJS) or Python FastAPI; REST with an OpenAPI spec; background job queue for notifications, scheduled reports and AI calls
Database	PostgreSQL with tenant and site IDs on every record; file storage for evidence uploads (S3-compatible)
AI	One LLM adapter module; JSON-schema outputs; embeddings + vector search (pgvector) for the Quality assistant
Charts	Recharts or ECharts
Exports	PDF (server-side render, hospital branding from Settings) and Excel
Auth	Email + password with password reset; role-based permissions; SSO shown as a connector, not built
Email	Transactional email through SMTP (a real mailbox for the sandbox, so notification emails actually arrive)
Hosting	Docker Compose; one cloud VM; HTTPS on an Antier subdomain such as app.healthcare-suite.antiers.work (sandbox tenant)
Connectors	HIS, WhatsApp and Google run through sandbox connectors behind one interface each, so a real connector replaces them later without UI changes

Non-functional:

- Every create, edit, AI suggestion and status change goes into the audit trail (who, what, when, before/after), viewable in Admin.
- An admin-only "Reset sandbox data" action restores seed data in under a minute, so each walkthrough starts clean.

- Fast on a laptop over a video call: pages under 2 seconds, no console errors, no placeholder text.
- Arabic and English UI labels via an i18n file; at minimum the patient-facing screens and messages must be fully bilingual.
- Version shown in the footer and an "About" dialog; favicon, page titles and email sender name all say Antier Healthcare Suite.

Out of scope, and never on screen

Out of scope for this build:

- Live integration with any HIS or EMR (including Sage), the WhatsApp Business API or the Google Business Profile API. Connector cards and sandbox connectors only.
- Appointment booking, scheduling, billing, claims or anything clinical (diagnosis, treatment, medication advice).
- Predictive no-show models, document AI on ID or insurance cards, computer vision. These stay in the presentation, not this build.
- A second populated tenant, working SSO, native mobile apps.

Never on screen, in code or in data:

- Any real hospital's name, logo or staff names, including the prospect's.
- Real JCI or Oman Ministry of Health standard text, or claims of certification.
- Real patient data of any kind; fictional names only.
- The words "demo", "POC", "prototype", "test" or "mock" in the UI, emails or URLs. The environment is marked only by the "Sandbox" badge.
- Fake customer logos, testimonials, "trusted by" lists, invented release history or "live at X hospitals" counters.

Build plan and acceptance

Plan on about 15 working days for the must-haves with one full-stack developer plus part-time UI design, and about 5 more for the should-haves. The developer should confirm this before we fix a walkthrough date; book it only after the checklist passes and two internal dry runs are done.

1. **Days 1–3: Product shell.** Design system and branding, auth and password reset, users, roles and permissions, organisation setup, navigation and global search, notification centre and email, audit trail, Settings, seed script.
2. **Days 4–6: Module 1.** HIS sandbox connector and event console, WhatsApp phone preview, bilingual feedback form, routing rule, feedback inbox, AI feedback analysis, cases with SLA and escalation.
3. **Days 7–10: Module 2.** Incidents with AI assist, investigation and corrective actions; infection-control audit templates and scheduling; accreditation tracker with evidence uploads; KPI dashboard.
4. **Days 11–12: Reports, integrations, assistant.** Saved reports with PDF/Excel and scheduled email, Integrations page, Module 3 enquiries, repeat-issue finder, Quality assistant over records and SOPs.

5. **Days 13–15: Polish and release.** Arabic/RTL pass, empty, loading and error states, performance, deploy to the sandbox URL, two internal dry runs.
6. **Days 16–20 (should-haves).** Document control, risk register, CSV import, privacy settings, Help centre.

Acceptance checklist:

- ☐ The 9-step storyline runs end to end, twice in a row, with no dead clicks or errors
- ☐ Every menu item opens a working page with data or a guided empty state; no "coming soon"
- ☐ An Arabic complaint is analysed live with translation, themes and urgency in under 5 seconds
- ☐ A 5-star response shows the Google review invitation; a 2-star response opens a case and never shows it
- ☐ An SLA breach raises an in-app notification, sends a real email and escalates to the Quality Manager
- ☐ A complaint can be turned into an incident with pre-filled fields, and the link shows on both records
- ☐ A new user can be invited, given a role and department, and sees only what that role allows
- ☐ Audit compliance %, accreditation readiness % and all KPIs calculate from the data, not hard-coded
- ☐ Settings changes (a theme, a KPI target, an audit item, the hospital logo) show up immediately across screens and exports
- ☐ The monthly quality pack exports to PDF with the hospital's branding from Settings
- ☐ The Quality assistant answers "top 3 repeat issues this month" with links, and refuses a clinical question
- ☐ Reset sandbox data restores seed data in under a minute
- ☐ No client name, real standard text, or demo/POC wording anywhere in UI, emails, code, data or URL
- ☐ Works on Chrome at 1440 px, and the patient form works on a phone

Open questions for the developer: the LLM provider and its Arabic quality; whether 10 days holds; and who writes the SOP sample documents (suggest Antier pre-sales).